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Phase change materials for enhanced photovoltaic panels performance: A comprehensive review and critical analysis

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ABSTRACT

Due to the high consumption of fossil fuels and the environmental challenges the world is facing, the transition towards renewable energies has become increasingly evident. Thus, enhancing the performance of different renewable energy systems became a fundamental research focus. Improving the annual solar photovoltaic systems efficiencies became a critical need to maintain system performance and durability. The major challenge facing solar photovoltaic system technology to be controlled and reduced is the overheating of the solar cells, where this factor not only affects panel efficiency but also causes heat-induced structural damages, and long-term exposure which may accelerate material degradation, reduce panel lifespan, and potentially develop hot spots. Among the different solutions is the use of phase change materials. This research demonstrates detailed recent literature review alongside with the appropriate classifications and critical analysis related to four distinct PCM-based cooling systems: pure PCM, composite PCM, finned PCM, and hybrid PCM systems. These PCM systems affect solar system efficiency, electrical power generation, and temperature. Findings revealed that hydrated salt HS36 and paraffin wax RT42 in pure PCM systems can highly enhance system electrical efficiency, as well as enhancements achieved through composite PCM systems incorporating multiwall carbon, graphene nanoplatelets, and magnesium oxide. Moreover, finned PCM integrated with zinc nanoparticles and aluminum fins, demonstrated promising efficiency improvements. On the other hand, hybrid PCM systems, such as PVT-RT35HC integrated with graphene nanoparticle nanofluids, show significant efficiency gains and electrical power enhancements, knowing that the majority of studies performed during the last three years were introducing hybrid cooling systems integrated with phase change materials. This research article shows the potential of PCM-based cooling solutions in advancing renewable energy technologies and covers a comprehensive review that goes through the recent studies of the last three years about employing phase-change materials as a cooling system for solar cells through different ways that can be classified into four main categories which are pure PCM, composite PCM, finned PCM and hybrid cooling systems.

1. Introduction

Millions of academics are increasingly prioritizing energy research for its numerous uses in varied daily activities, improving innovation around the world [1–8]. The present worldwide trend in energy research is strongly aimed towards reducing energy use and carbon dioxide emissions which reflects a high commitment to environmentally friendly and sustainable energy options [9–11]. To achieve these objectives, renewable energy sources must be harvested together with effective energy management strategies [12,13]. Waste heat recovery is an increasingly important procedure in the field of energy management. This approach has gained popularity in recent years as a result of significant heat discharges from engineering systems that are constrained by the limits of the second law of thermodynamics. Waste heat recovery has two advantages: it decreases energy waste while also reducing environmental effects, making it a key component of the current push for sustainable energy solutions [14–17]. Moreover, during the world COVID-19 and the Russian-Ukrainian war, it was found that it is critical to search for alternatives to fossil fuels by targeting renewable energies at the same time. When facing a shortage of fossil fuels, it can be seen how slow

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Nomenclature

AC	alternating current
$C_{27}H_{56}$	heptacosane
$CaCl_2.6H_2O$	calcium chloride hexahydrate
CFD	computational fluid dynamics
COP	coefficient of performance
CPV	concentrator photovoltaics
Cu	copper
DC	direct current
HS	hydrated salt
LHS	latent heat storage
NPCM	nano phase change material
OM29	organic matter
p-n	positive-negative
PCM	phase change material
PEG	polyethylene glycol
PT	PureTemp
PV	photovoltaic
PVT	photovoltaic-thermal hybrid solar collector
Q	flow rate
RT	RUBITHERM
SiC	silicon carbide
SP	self-priming model
TEG	thermo-electric generators
TES	thermal energy storage
ZnO	zinc oxide

is the rate when moving to renewable energies, which requires serious actions to be taken to improve the role and efficiency of renewables. The COVID-19 crisis had a negative effect on the development of the renewable, which is critical to take into consideration. In addition, current energy security is already in danger due to the Russian-Ukrainian crisis. Russia is the second-largest natural gas and oil producer in the world after the United States; it supplies 13 % of global oil and 17 % of global gas in 2020, where it provides more than 40 % of the Europe's gas [18]. In 2020, the total energy consumption was supplied by fossil fuels, renewable energies and nuclear energy, which were distributed as 83 %, 12.6 %, and 6.3 %, respectively. However, if global energy demand increases up to 50 % by 2050, the world can reach its fossil fuel independence by increasing renewable energy output by six or eight times. To reach this point, energy improvements that are being worked on related to renewable energies must be enhanced by 1.5 % more per year [19].

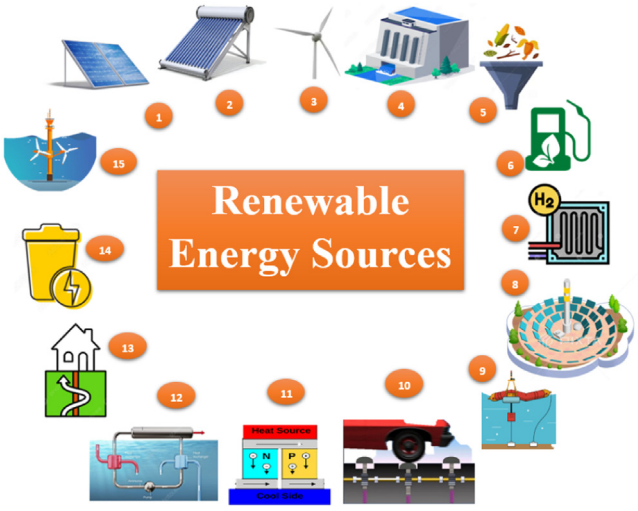


Fig. 1. Renewable energy sources.

1.1. Renewable energies overview

Renewable energy sources are always being recovered by nature and can be used for endless lifetimes. Moreover, the future is now moving toward renewables to avoid climate change issues and minimize greenhouse effects by employing renewable energies. However, renewable energies still face some challenges related to their usage criteria since they are highly dependent on the location where they are employed, on the weather conditions, and always need enhancements and technological improvements. To maintain the continuity of benefiting from renewables, continuous research studies for their improvements are required. Combining different sources together can lead to beneficial results. The main sources of renewable energy are shown in Fig. 1.

Moreover, Table 1 summarizes different recent research that discusses each of the mentioned renewable energy sources, with a brief definition for each of them based on their corresponding numbers in Fig. 1.

Current summaries of renewable energy have highlighted several obstacles that these various renewable energy sources may encounter, including climate change. Since some renewable energy sources, like biomass energy, depend on natural circumstances, there is a reciprocal relationship between renewable energy sources and climate change [24]. Furthermore, the intermittent nature of renewable energies poses

Table 1
Sources of renewable energy: definitions and corresponding recent researches.

Corresponding Source	Definition	RefS.
1. Solar Photovoltaic	Generates electricity from sunlight through solar panels.	[20]
2. Solar Thermal	Produces steam through fluid heating for driving a generator to produce electricity or for heating purpose.	[21]
3. Wind Energy	Wind turbines produces electricity when turning due to wind's kinetic energy.	[22]
4. Hydroelectric Power	Through dams or water turbines, flowing water's energy is harnessed to produce electricity.	[23]
5. Biomass Energy	Heat or electricity is produced from organic materials (wood, waste, etc.).	[24]
6. Biofuels	Electricity is produced through liquid fuels produced from organic matter.	[25]
7. Hydrogen Fuel Cells	Through chemical processes, hydrogen and oxygen are converted into electricity and water. (Hydrogen might be assured from renewable sources).	[26]
8. Concentrated Solar	Mirrors and lenses concentrate sunlight into smaller areas to produce electricity through conventional steam or gas turbines.	[27]
9. Wave Energy	Ocean's wave motion is harnessed to generate electricity	[28]
10. Piezoelectric Energy	Mechanical stress produced by certain materials when subjected to pressure or vibration is converted to electricity.	[29]
11Thermoelectric Generators	Employs See Beck effect, which converts temperature difference between two surfaces into electricity.	[30]
12. Ocean Thermal Energy	Power is generated through the use of temperature difference between ocean's surface and deep water	[31]
13. Geothermal Energy	Taps into the heat generated from the earth's core by usually pumping hot water or steam from underground reservoirs for heating uses or power generation	[32]
14. Waste-to-Energy	Solid wastes are converted into heat or electricity through different processes.	[33]
15. Tidal Energy	Makes uses of energy from rising and falling tides.	[34]

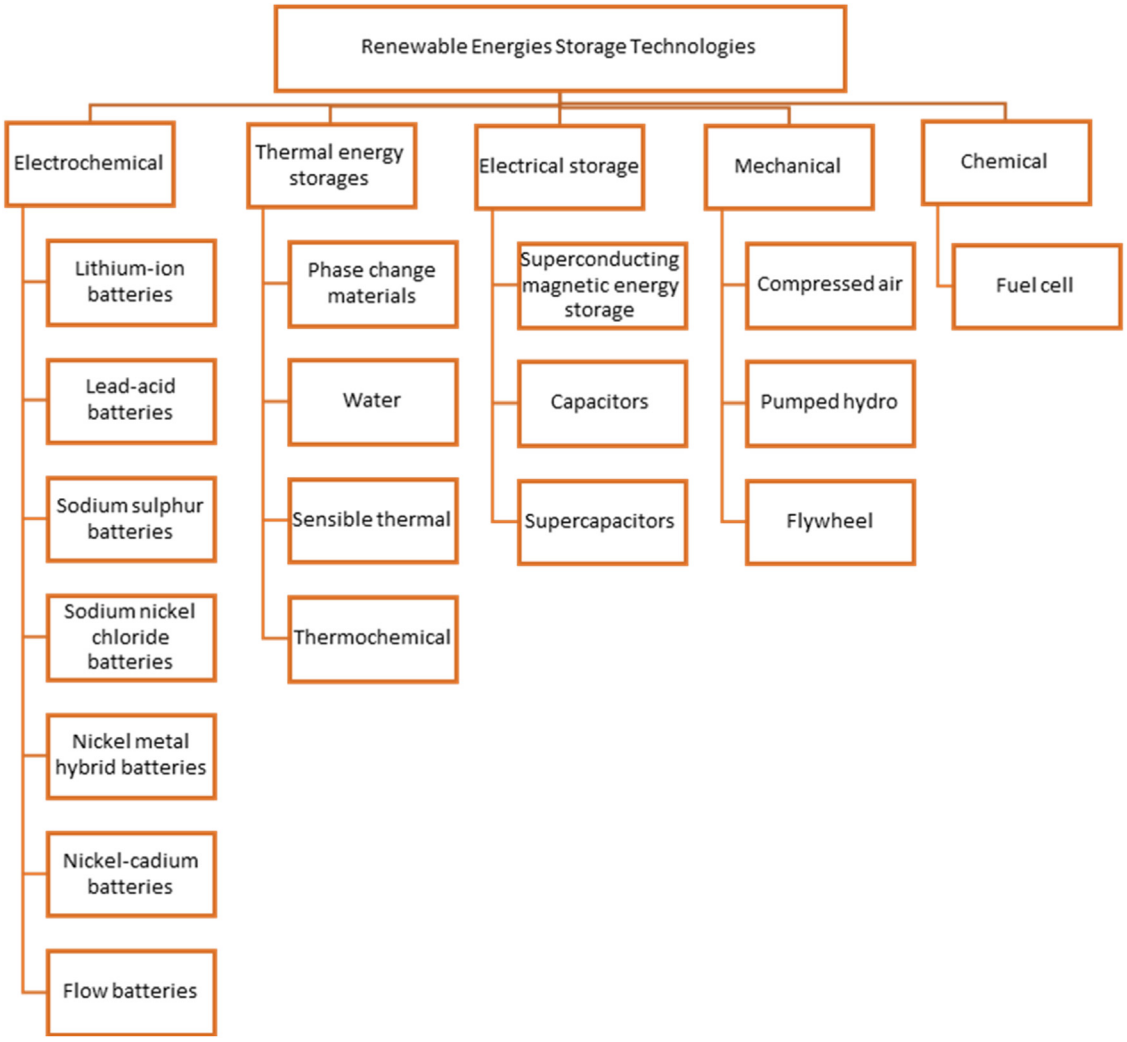


Fig. 2. Recent renewable energies storage technologies.

a significant issue. For instance, there are times when the wind blows and the sun shines, which has an impact on the process of producing energy. Energy storage devices were therefore the ideal way to store extra energy produced during times of high renewable energy production and release it during times of high demand or low renewable energy generation. In Fig. 2, many types of modern storage systems for renewable energies are grouped and displayed.

An analysis of the current sources of renewable energy highlights the benefits they provide, emphasizing their infinite supply and the global trend toward them as a result of climate change concerns. It was important to point out the limitations of renewable energy sources, such as their dependence on weather and geographic conditions, in order to emphasize the ongoing need for technological advancements. A summary of current research on various renewable energy sources is given in Table 1 and Fig. 1. In addition, Fig. 2's energy storage systems showed how to take the initiative to solve intermittency issues and maximize energy use.

1.2. Solar photovoltaics and factors affecting their efficiencies

Among all the renewable energies presented in Table 1, solar energy, specifically solar photovoltaics, is the cleanest and safest technology to generate electricity at small and large scales [35]. The photovoltaic effect turns light energy directly into electricity in a photovoltaic (PV) cell through layers of semiconducting material that are included within each cell. It is a type of photoelectric cell that can create and sustain direct current (DC) electricity that varies with the intensity of the sunlight when exposed to light without being connected to any external voltage source. In practice, this frequently necessitates conversion to certain needed voltages or alternating current (AC) via the use of inverters. Inside a solar module, many solar cells are linked. Modules are joined together to form arrays, which are then connected to an inverter, which provides electricity at the specified voltage and frequency/phase for AC [36]. A photovoltaic panel, as shown in Fig. 3. is made up of positively and negatively charged solar cells that work together to form a p-n junction diode. When incoming light breaks the junction's thermal equilibrium, free electrons in the depletion zone can quickly reach the junction's n-type side. Depletion holes can also quickly reach the p-type side of the connection. When a charge is placed across the junction, electrons migrate from the p-type side to the n-type side and vice versa, and current flows across the circuit. A typical single-junction silicon solar cell has an open circuit voltage of around 0.5 to 0.6 V; by combining them in a large solar panel, considerable amounts of renewable energy may be produced [37].

PV cells are subjected to different factors that affect their performance, which are presented in Table 2.

To keep PV performance at its best, a variety of issues and difficulties must be taken into account. In Table 2, the variables influencing PV performance were listed. The first element, solar irradiance, has a favorable impact on PV efficiency. Various factors, including location,

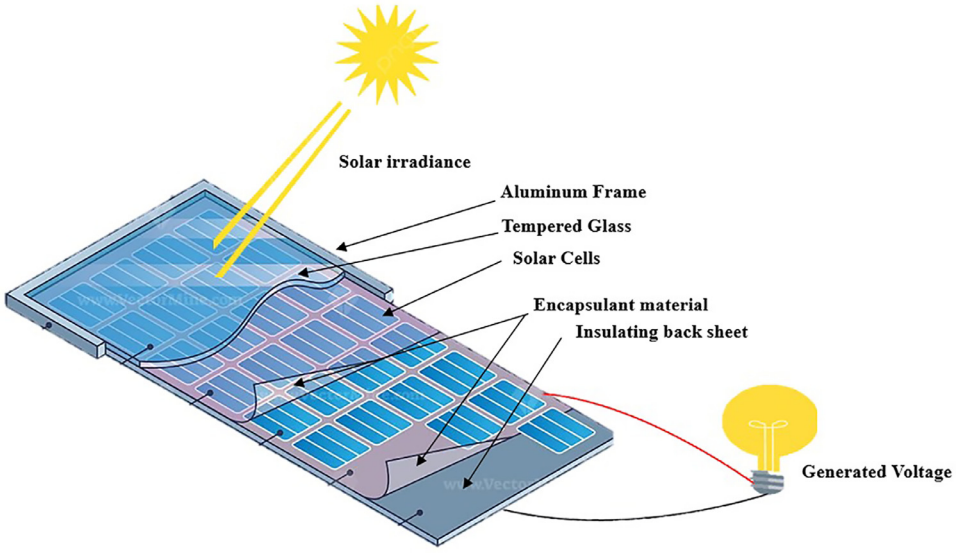


Fig. 3. Photovoltaic cell working principle.

Table 2
Different factors that affect PV cell efficiency, including their effect, suggested solutions and related reference.

Factor affecting PV cell efficiency	Effect of the factor and previous study results	Suggested solution	Previous study Refs.
Solar irradiance	Simulated PSIM monocrystalline and polycrystalline PV panels increase as sun irradiation increases.	Higher solar irradiation for higher efficiencies.	[38]
Shadings	the output power decreased by 41 % at 25 % shaded panel, and the power of the module was zero in the total hard shading situation. Moreover, the power was lowered by 57 % when one cell was shadowed.	Solar system location is critical to collect the most electricity from the modules.	[39]
Dust and dirt	The efficiency of a PV panel is proportional to its cleanliness. Light and heat influence the panel, and the presence of dust raises the PV surface temperature, increasing the power decrease rate. Dust accumulation reduces the power output by 8.80 % and the efficiency by 11.86 %	An effective cleaning process to recover the energy wasted. Water, detergent, and liquid soap all contribute to a 51 %, 60.1 %, and 68.4 % reduction in the power reduction rate of natural dust, respectively.	[40,41]
Rainfall and moisture	Humidity is a weather element that influences solar panel energy production. It has a less effect than temperature and solar irradiation.	Applying anti-reflective coatings to the surface of solar cells, such as silicon nitride or titanium dioxide.	[42]
5.Ambient temperature	Higher cell temperatures result in decreased power output, thus decreased cell efficiency.	Applying different cooling systems that are mentioned in upcoming sections to lower PV surface temperatures.	[43]

tilt angle, azimuth angle, etc., need to be taken into account for higher solar irradiance. In order to prevent shadings, which have a detrimental effect on PV performance and provide a substantial problem, these aspects must also be taken into account. Regarding dust and filth, several approaches, such as automated cleaning systems, could be investigated and implemented to lessen their detrimental effects. Further research on the durability and efficiency of coatings can be conducted to better understand the recommendations for applying anti-reflective coatings in relation to rainfall and moisture. The ambient temperature is the final factor that can be attributed to the decline in PV efficiency. Temperature control can be achieved through the use of various cooling systems; however, more research into cutting-edge cooling technologies may be able to address the difficult temperature conditions.

1.3. Overheating of solar cells

Solar photovoltaic systems are highly recommended as a renewable energy source for generating electricity. This innovative technology is an opportunity that must not be wasted and should be used correctly. However, with all their existing advantages, PV panels still face a major challenge, which is: overheating. Where, the ideal working temperature for solar panels is around 25 °C, when the temperature of these PV panels rises above this temperature of 25 °C which is considered the base line temperature, some sensitive components that are responsible for keeping the cell working properly begin to suffer from high temperatures and high heat absorbed. The overall performance of the PV panel is highly affected by high temperature, as a result, the efficiency of power and electricity production is also affected and will be decreasing significantly [44]. First of all, the solar cells, which are the main component of the photovoltaic panels, are the first component to be subjected to and highly affected by high temperatures and their stresses, which might cause structural damage and reduced electrical efficiency. Knowing that, electrical efficiency is the efficiency by which PV panels convert sunlight into power. Furthermore, the electrical efficiency is also affected when the materials that are responsible for transferring the electrons inside the PV panel are affected by high temperatures; their electrical conductivity will decrease. As a result, PV panels will not be able to optimally generate the required power. Moreover, high temperatures are directly related to what is called temperature coefficient, this coefficient reduces the effectiveness of solar panels. It reflects the rate at which the electrical output of a solar panel drops as its temperature rises. Beyond the technological challenges, less energy output from the photovoltaic system will result in extra economic and financial costs to cover the dissipated energy from another source [45].

Temperature increases in photovoltaic (PV) cells have two important effects: an immediate decrease in electrical performance (both in terms

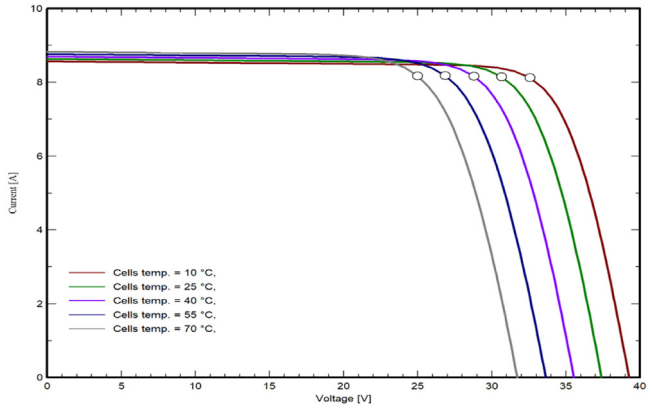


Fig. 4. Effect of temperature on solar cell voltage and current.

of output power and efficiency) and thermal stress, which significantly reduces the lifespan of PV systems. Furthermore, like other electrical components, silicon cells create heat, increasing the issue. Variations in operating temperatures, as well as isolated hot spots, cause power loss and permanent harm over time. Despite technical advances, only around 20 % of the incoming solar energy that strikes the cell's surface is turned into electricity, with the other 50 % wasted as excess heat. According to experimental data, every 1 °C increase in temperature reduces the relative electrical efficiency of PV modules by 0.4-0.5 %, as seen in Fig. 4. Elevated ambient temperatures can cause negatively charged stray currents to flow through the panel's integrated parts (such as the glass, mounting, and frame), resulting in silicon cell degradation and irreparable damage [46].

1.4. Various cooling systems

Controlling various parameters that affects solar photovoltaic systems is challenging; for example, solar radiation is an unpredictable factor in the design of photovoltaic panels on building facades, which are vertical and non-directional surfaces. However, controlling the overheating and panels temperatures is feasible, thus, enhancing efficiency by reducing the operating temperature is achievable. Thus, several cooling solutions has been carried out and assessed in a variety of different types of literature to make PV cells more efficient by avoiding the issue of temperature increase, as indicated in Table 3.

Table 3 presented a comprehensive overview of various available cooling systems to enhance PV performance, along with their summaries, results and references. As can be seen from the table, forced air cooling systems showed a rise in power as the number of DC fans employed at the panel's rear grew. Additionally, a drop in surface temperature and a noticeable improvement in system efficiency were noted by pushed water pipes, water spraying, and liquid immersion at various depths. Going on to phase change materials, whose precise thicknesses indicated a rise in system electrical efficiencies, may provide important information for PCM cooling system improvement. Moving on to radiative sky cooling, thermo-electric cooling, and heat fins, each of them showed a notable increase in efficiency. A noticeable marginal increase in electrical efficiency is observed for the spectrum filter cooling systems, indicating the necessity for additional system modification to yield more notable gains.

1.5. Phase change materials

There was an increase in interest in phase-change materials in recent years. PCMs have a unique capacity to absorb a substantial quantity of latent heat during the phase change process while incurring just a little temperature increase. Researchers have investigated the integration of PCMs with photovoltaic (PV) panels, resulting in PV-PCM modules. The major purpose of these modules is to keep PV cells running at lower temperatures, increasing conversion efficiency. When compared to typical PV/T systems, this passive cooling technology removes the need for fluids or extra power, minimizing maintenance requirements. Furthermore, PCMs are acknowledged as an efficient method of capturing thermal energy from renewable sources. An extensive study has performed

Table 3
Various available cooling systems with their related references.

Cooling System	System Summary	Previous Study results and effect on PV cell performance	Refs.
Forced air cooling system	Use DC directed fans at the back end of the panel. Where the power produced by the PV is proportional to the number of fans	With 1, 2, 3 and 4 fans used, the power increase by 12.93 %, 37.17 %, 41.28 % and 44.43 % respectively.	[47]
Forced water pipes	Water circulation inside a flat plate closed-loop pulsating heat pipe	Numerical investigation that showed increase in improvement rate by 23 % and 35 % for passive and active approaches respectively	[48]
Water spraying	Water is sprayed through the sprinklers on the front of the PV modules in this system	Increase in electrical power output about 33.3 %, 27.7 %, and 25.9 % by using the steady-spray water cooling and the pulsed-spray water cooling with DC = 1 and 0.2 respectively	[49]
Liquid immersion	PV is installed at various depths of water in which water is utilized as an immersed fluid. Water acts as a heat dissipator, keeping the surface temperature of the solar cell constant.	The immersion of the panels in water at a depth of 20 mm increases the PV efficiency by 9.1 % compared to the PV without water immersion, where the depth and efficiency are proportional	[50]
Phase change materials	Placing a phase change material (PCM) which is paraffin wax RT-42 at the panels back surface	At a PCM thickness of 3 cm, the electrical efficiency increases by 14.4 % using PCM cooling	[51]
heat fins	Consists of adding additional component like heat sinks, metallic materials such as fins installed on the back of P.V. to provide convective heat transfer from air to. High thermal conductivity heat sinks are often situated behind the solar cell and increase the area of heat transmission from the solar cell to the ambient environment	The CFD simulation in the heat sink model with an air flow velocity of 1.5 m/s and temperature of 35 °C under a heat flux of 1000 W/m2 indicated that the average temperature of the PV panel decreased from 85.3 °C to 72.8 °C.	[52]
Radiative sky cooling	Radiative cooling of PV enhances solar panel efficiency by emitting excess heat as infrared radiation, improving energy conversion.	Using a combined opto-electro-thermal simulation, a Nano micro-scale cooler was tested, resulting in an efficiency improvement of roughly 3.1 % and a drop in operating temperature.	[53]
Thermo-electric (Peltier) cooling	The thermoelectric module captures and converts surplus heat from the PV panel into electric power; this system is based on the Peltier effect, which is utilized to cool the PV panels using electrical energy.	Results showed temperature reduction using Peltier cooling, where an average temperature of 33.37 °C was recorded for against 45.60 °C for the cooled and uncooled panel respectively.	[54]
Spectrum filter cooling	These liquid spectrum filters are fitted on a thermally protected PV module. The filters transmit just the usable spectrum to the PV while filtering out the rest	With Silicone oil spectrum filter, the average electrical efficiency of the PV module was found increase marginally	[55]

to examine their use in solar energy and energy-saving in buildings. Despite rising interest in PV-PCM technology and significant studies in this field, its practical use remains uncertain due to PCM limitations: low thermal conductivity, limited heat storage capacity, potential leakage, and high pricing. As a result, current PV-PCM systems remain inadequate in terms of dependability and cost-effectiveness when compared to other solutions, where a complete assessment of the advances in PV-PCM research is necessary [56].

A previous review about cooling systems for PV cells that is based on phase change materials covered some previous works from 2003 until 2017 that employed PCM for cooling the PV panel in different methods, like pure PCM, composite PCM, finned PCM, and hybrid PVT/PCM with nanofluids [57]. Another review also takes into consideration the cooling systems for PV panels, including PCM. It took into details the different PVT and nanofluid cooling systems integrated with PCM as active cooling and get into passive cooling methods employing nano-PCM until 2020 [58]. In addition, another literature summarized the use of pure PCM in cooling PV panels and improved the PCM performance by adding fins, impurities, composite PCM, encapsulated PCM, and nano. However, the mentioned research also covered the researches until 2020 [59]. Also, some researches focused on the use of pure PCM in cooling PV systems and included their thermo-physical properties in details such as melting point, density, latent heat, thermal conductivity, and specific heat [60]. Moreover, challenges for cooling PV panels using PCM were also taken into consideration during the recent literature reviews [51].

What makes this review distinct from other review is particular consideration of the most significant experimental and numerical recent studies that were performed during only the last three years about the cooling of photovoltaic solar systems using phase change materials. The work on the last recent studies is critical since it will be the base for the researchers and scientists to build on and continue their studies since the technological advancements are increasing rapidly, and it was notable to take into account numerical investigations because the world is now directed through the world computational simulations. These tools should be beneficial and integrated with the recently used software to improve studies and advancement in a time and cost saving manner with significant results.

1.6. Review methodology

The main goal of this study is to evaluate the efficiency of solar systems, with a focus on controlling the overheating challenge through different cooling systems that might be employed, specifically Phase change materials. Starting from the environmental issues happening nowadays and the world's road toward renewable energies, then focusing on solar photovoltaic systems and the challenges they're facing during the research process, the main keywords that were employed to help and direct the research were: renewable energy, cooling systems, solar systems, photovoltaic cells, phase change materials, thermal energy storage systems, pure PCM, composite PCM, and hybrid systems. To cover the most recent updates and technologies in the field of using PCM as cooling systems for photovoltaic panels this study focused on covering the publications between 2020 and 2023. Furthermore, inclusion and exclusion criteria were followed guarantee that the research will maintain a high-quality. Articles that were published between 2020 and 2023 employing experimental or computational studies with outcomes related to energy or efficiency increasing and temperature decreasing of the system by employing PCM cooling technologies were included among the chosen articles. Moreover, chosen articles should include full specifications about the type of the used PCM, the technique and setup used, a system description, and comprehensive results and conclusions. The articles that did not meet these criteria were excluded from the research. Following that, obtained outcomes from researchers were analyzed and organized to create the state of the art of this recent review about solar photovoltaic systems with an emphasis on PCM-based as a cooling system for photovoltaic cells. The main methodology of the research is mentioned in Fig. 5.

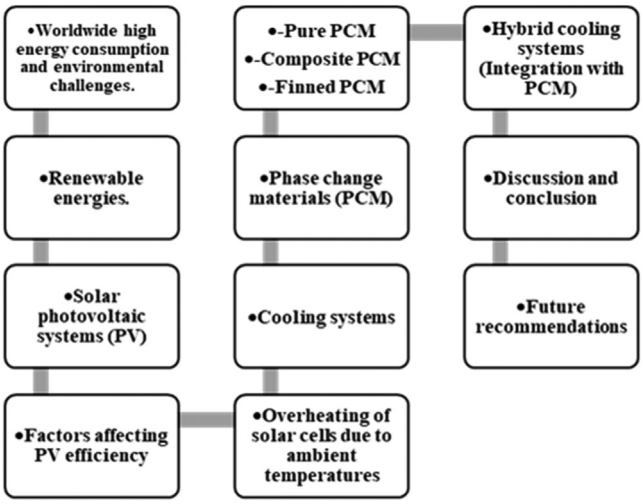


Fig. 5. . Research methodology.

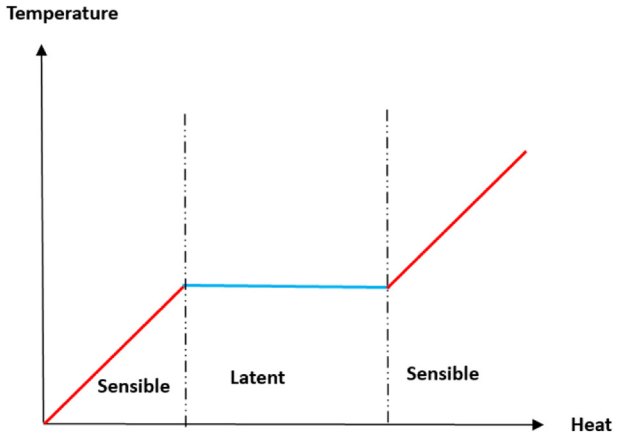


Fig. 6. Temperature control of sensible and latent heat during the phase change.

2. Theoretical background

2.1. Overview on phase change materials

Thermal energy storage systems are classified into three types which are sensible heat, latent heat, and reversible thermochemical reaction. Latent heat storage is the most important type of thermal energy storage, with a focusing on employing the phase-change materials as the latent heat storage system [61]. Phase-change materials are recently highly being used as thermal energy storage systems due to their high energy density, high temperature, and beneficial energy output stability [62]. A comparison of sensible and latent heat storage systems was done by employing water as sensible heat storage and paraffin as a phase change material for latent heat storage. The study indicated considerable performance differences. When both systems are compared employing the same storage medium volume and heat transfer fluid temperature, it was clear that the phase change material energy storage capacity exceeded the water energy storage capacity. The phase change material stored 4.1 times more energy than water while having less temperature variations in the storage medium during the phase change process [63]. Moreover, Fig. 6 demonstrates the control of temperature during the phase change process, highlighting the distinctions between sensible heat and latent heat regulation.

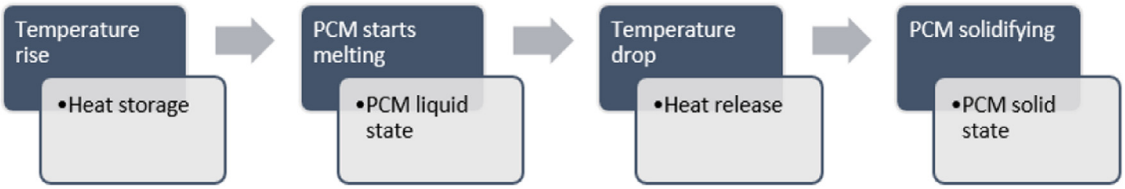


Fig. 7. PCM during energy storage and release process.

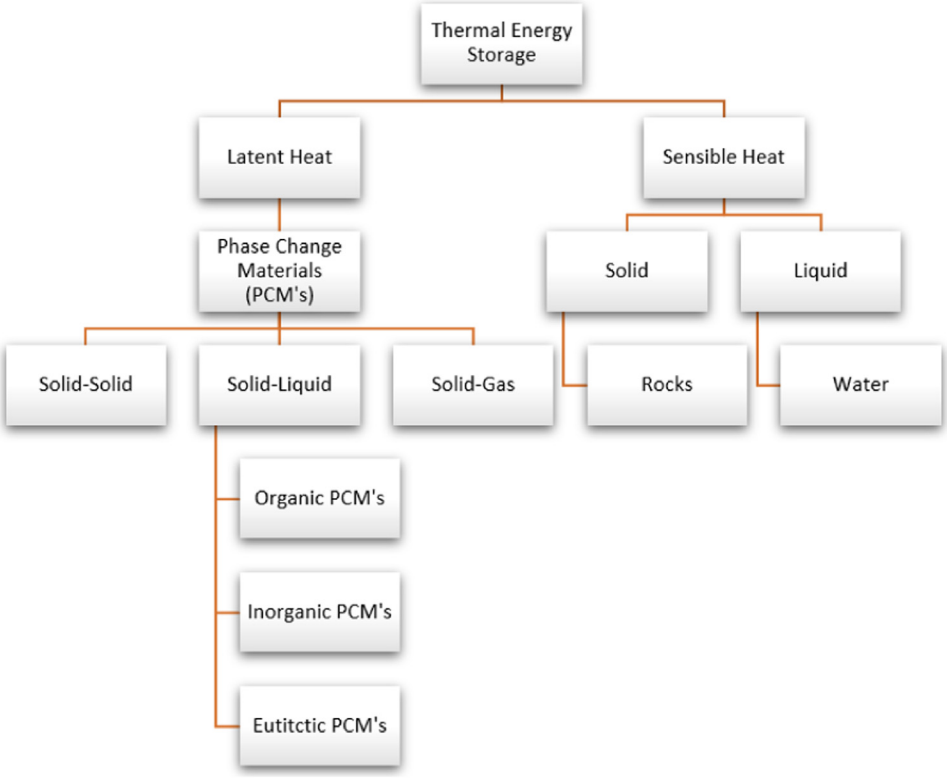


Fig. 8. Classification of thermal energy storage systems including examples.

Due to the mentioned reasons, latent thermal energy storage based on phase change material (PCM) has caught the interest of researchers in the disciplines of energy storage and usage [64]. Latent heat storage works on the idea that PCMs during their phase change process absorb and release huge amounts of latent heat [65]. During the PCM phase transition, a significant amount of energy can be absorbed or released within a short temperature range, generally confined to a few degrees Kelvin. Because of this property, the PCM may behave as a virtually isothermal heat reservoir [66]. The melting and solidification process of phase change material (PCM) which is shown in Fig. 6 is critical in thermal energy storage because it allows for effective heat absorption and release inside the PCM. (Fig. 7)

PCMs can be used to store energy in three different ways: solid-solid, solid-liquid, and solid-gas. Solid-liquid phase change materials (PCMs) are considered the most viable due to their minimal volume variation and significant latent heat density [65]. As seen in Fig. 8, there are numerous varieties of solid-liquid PCMs with different melting points that fall into three main categories: organics, inorganics, and eutectics [67].

More details for each type of Solid-Liquid PCM's are listed below:

- Organic PCM's: Organic phase change materials (PCMs) such as sugar alcohols, paraffins, and fatty acids have benefits in thermal energy storage systems. These benefits include reduced corrosiveness, which aids in the long-term integrity of storage components. It is essential to highlight, however, that these organic PCMs have negative aspects. One such constraint is the difficulty in obtaining phase separation within these materials. As a result, distinguishing between the solid and liquid phases during the melting and solidification process may be more challenging than with other types of PCMs. As a result, while organic PCMs offer advantages, resolving phase separation difficulties is a critical concern in their application for effective heat storage [68]. More details about organic PCM's advantages and disadvantages are listed in Table 4.
- Inorganic PCM's: Inorganic phase change materials (PCMs) are divided into two groups: salts and salt hydrates, and metals and their alloys. These materials have two times the heat storage capacity per unit volume, greater thermal conductivity, higher working temperatures, and are less expensive than organic PCMs. However, they have significant disadvantages, such as being corrosive to metals, resulting in a shorter system service life and higher costs. Furthermore, inorganic PCMs, particularly salts and salt hydrates, may experience phase separation and supercooling, both of which might reduce their energy storage capability. Metals and metallic alloys, on the other hand, do not have these drawbacks and are prospective PCMs for high-temperature applications [69].
- Eutectic PCM's: A eutectic phase change material (PCM) is a specific composition made up of a mixture of organic and inorganic components. These eutectic mixes have particular benefits due to their distinctive features, which include a well-defined and accurate melting point as well as a notably high latent fusion heat. Because of this accuracy in melting point, PCM transitions between solid and liq-

Table 4
Organic PCM's merits and demerits.

Organic PCM's Merits	Organic PCM's Demerits
It is not corrosive.	Enthalpy of phase transitions decreases
Chemically and thermally stable Modifiable transition zone	Thermal conductivity, density, and melting point are all lower.
Available in a wide variety of temperatures	Flammability
Has no segregation.	Higher volatility
Recyclable	Cost is excessive.
Has a higher nucleation rate	Volume changes during phase transitions

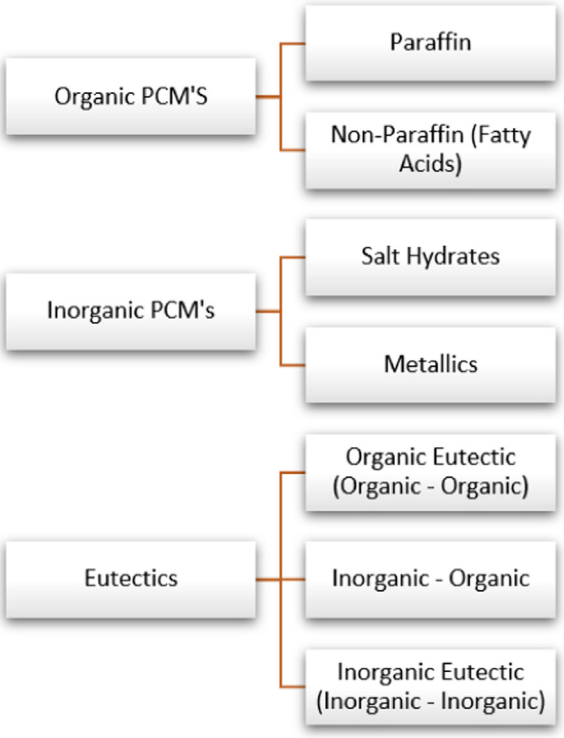


Fig. 9. Examples on organic, inorganic and eutectic PCM's.

uid phases at particular temperatures, making it highly dependable for a variety of thermal energy storage applications. Furthermore, because these materials have a high latent fusion heat, they may absorb or release a significant amount of energy during phase transitions, which contributes to their efficiency in controlling thermal energy. Eutectic PCMs have received a lot of interest in the fields of thermal energy storage and thermal management systems, where precise control and efficient energy consumption are critical [70].

Fig. 9 depicts many types of solid-liquid phase transition materials, as well as specific examples for each category.

Moving to the features of the ideal PCM used in latent heat storage, designing the proper phase change material encapsulation may be challenging, where parabolic and cubic shapes offer superior thermal management than linear profiles for the same volume of PCM. Moreover, consider the density of the phase change material while designing compact systems, since higher density improves thermal control. In addition, thermal conductivity is directly related to system performance and may be improved through several approaches. Similarly, latent heat is related to system performance. Finally, greater melting point temperatures result in better PCM performance.

The main properties for an ideal PCM to be used in latent heat storage [62] are shown in Fig. 10.

2.2. Applications of phase change materials

The introduction and integration of new renewable energy sources such as solar and wind power has critical in the goal of environmental sustainability and energy efficiency. Nonetheless, these green energy sources are necessarily unstable and variable. PCMs provide a critical link between energy generation and consumption, successfully minimizing the issues associated with renewable energy's unpredictable characteristics. Furthermore, PCMs have the unique capacity to maintain a steady temperature during the phase change, making them important in a variety of applications [71].

2.2.1. Solar water heaters

In order to increase solar water heater energy storage and thermal performance, several studies have identified latent heat storage (LHS) of phase change material (PCM) as an effective solution that has a high energy storage density and absorbs or releases heat at almost constant temperature. According to the findings, the efficiency of a solar collector with a PCM can potentially be raised by reducing energy losses from the storage tank to the ambient and allowing the system to be less affected by changes in solar radiation [72].

2.2.2. Concentrated solar systems

The use of phase change materials (PCM) in concentrated solar systems (CPV) can be highly effective, since the PCM is able to absorb heat generated by CPV cells and store it as latent heat without raising the actual temperature of the cells. As a result, the method might be employed to lower the temperature of the CPV cell, improving its power production and efficiency [73].

2.2.3. Cooling photovoltaic systems

To limit and manage the temperature of photovoltaic panels, the scientific community has devised a variety of novel solutions; one of them is passive cooling with phase change materials (PCM). This technology uses the unique thermal capabilities of PCMs to absorb and store excess heat generated by PV panels during operation, assisting in the maintenance of panel temperatures and improving efficiency and lifetime in a variety of climatic circumstances [74].

2.2.4. Solar greenhouse

When compared to ordinary greenhouses, the experimental greenhouse employing PCM revealed a considerable rise in ambient temperature (1–8 °C) after midnight [75].

2.2.5. PCM Trombe wall

Prior investigations showed that a Trombe wall composed of paraffin wax might keep the room warmer for about 9 h when compared to other materials. Furthermore, as compared to a typical Trombe wall, the PCM Trombe wall may reduce interior temperature fluctuation and indoor overheating in the summer [76].

2.2.6. PCM wallboards

By decreasing temperature swings, PCM wallboards increase interior comfort in both the cooling (summer) and heating (winter) seasons. Due to the varied solar impacts in each season, the PCM is more efficient in the winter than in the summer [77].

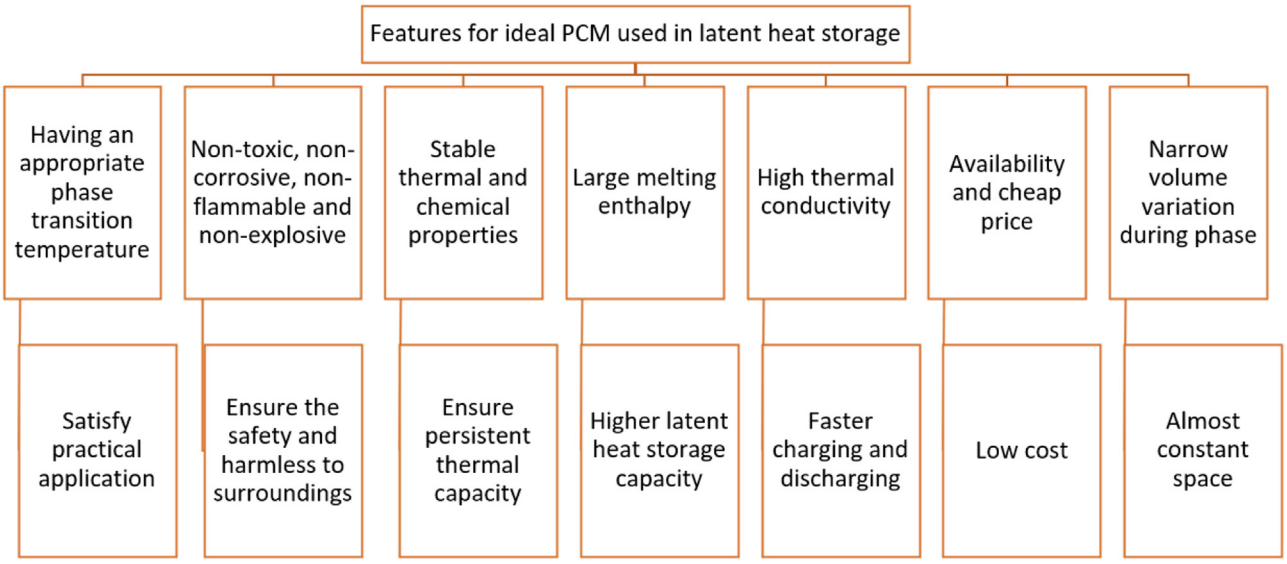


Fig. 10. Features for the ideal PCM used for latent heat storage systems.

2.2.7. PCM in heat recovery

PCM may be used for waste heat recovery, such as engine heat, where the heat generated from the PCM can be released and employed to generate steam, which can then be used for a variety of purposes. Engine studies revealed that the procedure without PCM produces more toxic gases that are dangerous to people and the environment, whereas the system with PCM produces the lowest exhaust gas temperature [78].

3. Cooling solar systems using phase change materials

Change of Phase Thermal regulation materials are gaining popularity in the field of photovoltaic solar cell technology. PCMs are chosen for their exceptional energy storage capabilities as well as their ability to perform regularly under constant temperature conditions. Researchers have thoroughly investigated the integration of PCMs with PV systems in order to improve their performance. For example, because of its wide temperature range, high heat of fusion, and ability to freeze without supercooling, paraffin is a popular PCM for this purpose. Experimental and numerical analyses yielded beneficial findings, revealing a 4.3–8.7 % improvement in power production. Furthermore, researchers used PCM to collect extra heat during sunny hours and then release it during non-sunny hours, leading to a 9 % increase in electrical efficiency. Furthermore, mixing PCMs with water and nanofluids has resulted in improved electrical efficiency, with PVT and PCM PV solar cells reaching up to 15.4 % efficiency. Furthermore, some PCM additions, such as fins or aluminum plates, were shown to enhance heat conductivity and electrical efficiency by up to 2 %. Incorporating a copper foam matrix, for example, boosted electrical efficiency by 5.68 %, while the inclusion of nanoparticles improved electrical performance by 6.8 % to 12.1 % [79].

Different methods using PCM for PV panels cooling are presented in Fig. 11.

Simple PCMs constitute the basis of employing PCMs to cool PV systems. They provide effective thermal management by collecting and discharging heat during transitions between phases. Moving a step farther, by the use of composite PCMs through mixing materials with various phase change temperatures. This improves PV systems' flexibility to a variety of environmental circumstances. In addition, Finned PCMs and configurations with two PCMs separated by fins improve heat transmission even more, improving the overall efficiency of PV systems with PCM. Also, to attain even higher efficiency benefits, PCMs are used in hybrid photovoltaic thermal (PVT) systems by water- or air-based

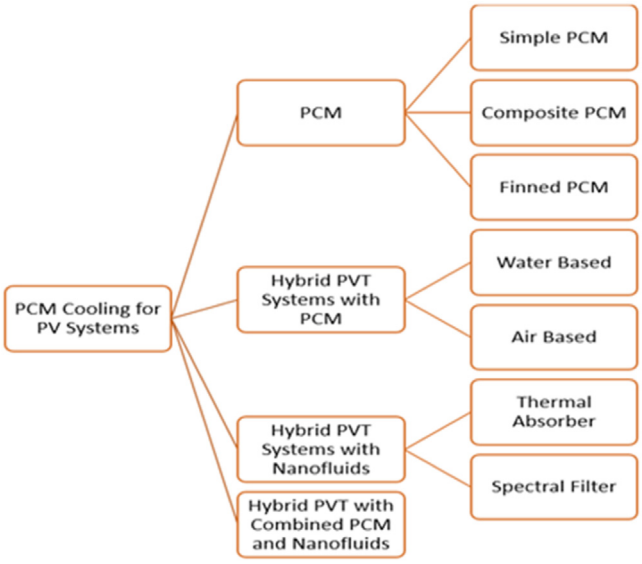


Fig. 11. Different methods of cooling PV systems using PCM's.

systems. Water is used as a heat transfer fluid in water-based systems, whereas air is used as the thermal medium in air-based systems. Another area for improvement is the application of nanofluids in hybrid PVT systems. Nanofluids act as both heat absorbers and spectrum filters in this approach. They absorb solar energy effectively and allow for adjustable spectrum selectivity. Because of the improvements in thermal and electrical performance, hybrid PVT systems employing nanofluids are interesting alternatives for more efficient solar energy conversion. Researchers have also investigated the complementary advantages of mixing PCMs and nanofluids in hybrid PVT systems. These systems achieve greater heat management, energy collection, and overall efficiency by combining both methods.

3.1. Cooling PV panels using simple/pure PCM

The cooling of the solar PV systems is essential during hot climates in order to keep the system's performance as intended so their efficiency do not drop due to high temperature and overheating. Where

pure phase change materials (PCMs) can be a suitable cooling system, such as paraffin waxes, they provide many advantages when employed for cooling the solar cell. The PCM works on the principle of collecting heat from the photovoltaic cells during high temperatures (most of the time is during peak sun hours of the day). During this collection, they transition from a solid to a liquid state at a steady temperature, allowing the PCM to manage the operating temperature in the PV panels to prevent them from overheating. Pure PCMs do not separate phases or deteriorate during their phase change, which is typically the case when talking about composite PCMs [80]. Furthermore, pure PCMs have exceptional thermal stability, which allows them to collect and discharge thermal energy without losing their performance efficiency. This simple and dependable cooling method improves the overall energy production of solar systems, thus making financial savings. That's why pure PCMs are a viable and long-term option for cooling photovoltaic [60].

Moreover, Table 5 summarizes the research that are done between 2020 and 2023 about the usage of pure PCM for cooling solar systems. These studies look investigated different PCMs, including the study methodology, system configurations, and the results of the cooling system on the panel's performance, efficiency, and operating temperature. The cooling properties of paraffin wax, hydrated salt, polyethylene glycol, soybean wax, and other PCMs were investigated when they were attached to the rear surfaces of solar panels, showing major decreases in operational temperatures and improvements in the electrical efficiency. Furthermore, it was shown that PCM thickness and panel tilt angles are important factors in maximizing solar panel performance since they can be employed in a beneficial configuration regarding potential solar system design and deployment in a variety of climatic circumstances.

3.2. Cooling PV panels using composite PCM

In the field of PV panel cooling, composite PCMs have distinct advantages. These materials integrate the heat-absorbing characteristics of typical PCMs with additional improvements like nanoparticles or fibers. By adding these nanoscale elements into the PCM matrix, the transfer of heat and thermal conductivity properties of the composite PCM may be greatly improved. This implies that when the PCM accumulates heat and transforms from a solid to a liquid phase throughout the phase change procedure, it can effectively disperse this heat. As a consequence, the PV panels remain cooler over extended periods of time, guaranteeing consistent and ideal working temperatures. This longer cooling effect given by composite PCMs adds to increased energy yield and PV panel longevity, thus enhancing their total value [89].

Table 6 shown below summarizes the most relevant research done between 2020 and 2023 that had an emphasis on the usage of composite phase change materials (PCMs) as a cooling system for PV panels, the summary took into consideration the PCM formulations and their combination methods employed to improve the performance and efficiency of PV panels. The main objective of these studies was to reduce excessive heat and increase energy output through the implementation of phase change materials on the back surface of PV panels. And the results obtained were also mentioned in the table, showing a temperature decreases of up to 20 °C and an improvement in electrical efficiency of up to 46.11 %. Furthermore, the integration of nanoparticles has a big role on improving PV performance, such as zinc oxide, alumina, silica, copper, graphite, and titanium were very beneficial, and this enhanced the PCM performance, thus enhancing the PV cells performance.

3.3. Cooling PV panels using finned PCM

Cooling solar systems through PCMs and finned heat exchangers is a very efficient and ecological alternative, where PCMs are used with finned thermal exchangers. Usually, PCMs function as a thermal battery that holds surplus heat created by solar cells during maximum sun hours and unleashed it when needed delivering a continuous and steady energy supply. When adding fins, this improves heat transmission efficiency since they enhance the amount of surface available for the exchange of heat, and allowing the PCM to be charged and discharged more quickly. This combination increases total solar system performance by maintaining ideal operating temperatures and lowering the danger of overheating, which can lengthen the useful life of solar panels and boost their energy production [96]. Moreover, Table 7 summarizes a number of current research undertaken between 2020 and 2023 with the aim of improving the cooling performance of solar systems by employing finned phase change materials (PCMs). Studies show a variety of techniques, configurations of systems, and PCM options where researchers have routinely obtained remarkable outcomes using software models and experimental settings. Using graphite and copper fins instead of steel and titanium, for example, resulted in a significant decrease in the solar cell temperatures, and the PV cell temperatures were reduced by 32 K with the addition of PCM films with aluminum rectangular fins. Furthermore, the combination of PCM and nanoparticles, as well as the use of low-cost aluminum reflectors, revealed enhanced electrical power production and lower temperatures.

3.4. Hybrid PVT systems with PCM

The combination of PCM with hybrid cooling systems is necessary to maximize thermal control in photovoltaic systems. PCMs provide a unique set of benefits mentioned earlier in the previous section. Furthermore, the usage of PCMs in hybrid cooling mechanisms results in significant energy and cost savings in different domains like solar power, constructions and transport. These systems focus on lowering total energy consumption and contributing towards an environmentally friendly world [57].

Table 8 summarizes recent studies and improvements in combining phase change materials (PCMs) inside hybrid and various cooling systems for solar power panels, where with the different advantages of PCM, they can improve the efficiency and effectiveness of solar energy systems under a variety of environmental circumstances. Researchers have investigated several PCM kinds and combinations in the search for sustainable energy solutions; for example, the usage of encapsulated PCM balls in conjunction with active and passive cooling techniques resulted in dramatically increased thermal and electrical energy gains. When paired with micro fin pipes, nano PCMs and nanofluids demonstrated significant temperature reductions and improved thermal efficiency. Furthermore, the incorporation of PCMs alongside heat pipes, porous media, and thermoelectric generators has regularly shown significant improvements in PV panel efficiency and electrical output, highlighting the important role of PCM-based combined cooling systems in advancing renewable energy technologies. These studies collectively highlight the importance of PCM integration in the development of effective and environmentally friendly solar energy systems. For example, on hybrid cooling systems employing PCM. Previously, a cooling system using PCM that have used in conjunction alongside a cooling nanofluid flow system to manage the system's heat capacitance. Nano-SiC particles were also employed in paraffin-PCM with nanofluid [103]. The design was tested in the open air at a fluid flow rate of 0.17 kg/s. The suggested solution decreased cell temperature (30 °C) during the peak of solar irradiation time (12:30 AM to 1:30 PM), which improved PV/T system performance and increased power from 61.1 W to 120.7 W, as well as improved the electrical efficiency from 7.1 % to 13.7 %. Furthermore, the system maximized the usage of the thermal energy that was gained, with a thermal efficiency of 72 % [103].

Another example of hybrid cooling systems combined with PCM is the notion of a combined thermoelectric PCM system to improve PV ef-

Table 5
Recent studies ranging between 2020 and 2023 on cooling solar systems using pure phase change materials, with the method of study, system configuration and setup, results obtained and a final conclusion.

Refs.	Year	PCM Used	Methods	Configuration	Results	Conclusion and notes
[81]	2023	Paraffin Wax (RT-42)	Experimental	Attaching PCM to the back surface of the panel, studies at different tilt angles and PCM thicknesses	At 3 cm thickness and 30° tilt angle, increase of 15.8 % and 14.4 % in electrical power and electrical efficiency respectively	Electrical power and efficiency are proportional to PCM thickness (up to 3 cm) and to tilt angle (up to 30°)
[82]	2023	Hydrated salt (HS36) with latent heat 167 kJ/kg	Experimental	An experimental setup was designed and installed according to the latitude position at a tilt angle of 11.5° at the top of an institute.	The panel's operational temperature was decreased by 25.4 % which was 16.98 °C. while the electrical efficiency of the panel increased by 17.5 %.	With the integration of PCM with different encapsulating materials, the study may be extended with other types of PCMs for varied climatic situations.
[83]	2022	PEG - Polyethylene glycol 1000 (melting temperature is 33–39 °C)	Experimental	Using PCM to absorb additional heat from the rear of solar panels, which allows for greater energy efficiency	Increases in the system's electrical efficiency by 4.82 %	PCM are useful temperature controllers for PV solar cells.
[84]	2022	Soybean wax	Software Simulation (Solar Simulator)	50 Wp polycrystalline solar cells combined with and without soybean wax were installed on the backplate of solar panels utilizing a PCM container to serve as a passive cooling system.	Reduction in optimum Photovoltaic temperature during 1100 W/m2 intensity from 60.7 °C to 54.7 °C and improvement in maximum efficiency at 900 W/m2 intensity by 0.42 %	The soybean wax cooling method have shown to be successful in lowering PV temperature.
[85]	2021	Polyethylene glycol 600 (PEG-600)	Experimental	PEG-600 melting temperatures ranging from 23 to 26 °C, it is colorless, non-toxic, odorless, and non-acidic. And placed on the backside of 60 W PV panel.	Temperature drops by 18.3 and efficiency increase by 2.45 %	It is not suggested to use materials with low melting points in places with high ambient temperatures.
[85]	2021	Paraffin	Experimental	melting temperatures ranging 56–58 °C. PCM is colorless, non-toxic, odorless, and non-acidic. placed on the backside of 60 W PV panel	Temperature drops by 40.2 °C and efficiency increase by 2.8 %	It is not suggested to use materials with low melting points in places with high ambient temperatures.
[86]	2020	RT-35HC (RUBITHERM) - Organic Paraffin PCM	Experimental	Experiments were carried out from 9 a.m. to 4 p.m. on days with almost identical ambient conditions in July. 30 W monocrystalline PV panels were utilized. To gather the most sunrays, the tilt angle of the solar panel was set at around 34° (latitude angle: 33.7).	Temperature dropped by 11.9 °C and electrical efficiency increased by 9.1 %	The usage of nanofluid in conjunction with PCM improves system performance over PCM alone. Furthermore, a combination system of PCM and nanofluid is more effective than the integrated PCM and water system.
[87]	2020	OM29	Experimental	To overcome the conduction barrier, liquid PCM is placed directly on the rear surface of the PV module, and PCM obtains heat directly from the PV module. Furthermore, the back face of PCM is sealed by tin and aluminum to prevent leaks during phase shift.	OM29 reduced PV module temperature by an average of 1.2 °C till 08:30. The OM29 PCM, however, was unable to decrease it after 09:00.	The reduced melting point of the OM29 was linked to the given PCM's failure to reduce the operating temperature of the photovoltaic panel.
[88]	2020	Calcium Chloride Hexahydrate ($CaCl_2{\cdot}6H_2O$)	Experimental with TRNSYS software simulation	PV panels were positioned in "Sandwich model" for the PV-PCM panel. Two translucent plexiglass sheets with a thickness of 8 mm were held together by 5 mm screws.	The highest temperature decrease is 26.3 C (38 % reduction) when compared to the traditional PV module. With power output increase by 1.16 W (24.68 % increase).	The largest discrepancy between actual and simulation values was 0.96 % and 8.25 %, indicating high agreement among these data.

Table 6
Recent studies ranging between 2020 and 2023 on cooling solar systems using composite phase change materials, with the method of study, system configuration and setup, results obtained and a final conclusion.

Refs.	Year	PCM Used	Methods	Configuration	Results	Conclusion and notes
[90]	2023	Paraffin was PCM mixed with nanoparticles ZnO (Zinc oxide)	Experimental	Incorporating PCM (Paraffin wax)/zinc oxide (ZnO) nanoparticles on the rear surface of a photovoltaic panel with capacity of 50 W	Reduction of 16.5 %. in temperature and increase of 12.18 % in output power compared to an uncooled panel	Nanoparticles can enhance PV performance when employed with PCM
[91]	2023	$CaCl_2{\cdot}6H_2O$	Experimental	Lowering the temperature of operation of the PV module's back surface	The average temperature of operation drops by up to 20 °C and 46.11 % greater average power output obtained with a 6 % better electrical efficiency when using PCM	By lowering the operating temperature of PV modules, phase change materials increase their energy efficiency as well as their power.
[83]	2022	Alumina nano PCM (Polyethylene glycol 1000)	Experimental	Using PCM to absorb additional heat from the rear of solar panels.	Increases in the system's electrical efficiency by 8.1 %	NPCM are useful temperature controllers for PV solar cells.
[83]	2022	Silica nano PCM (Polyethylene glycol 1000)	Experimental	Using PCM to absorb additional heat from the rear of solar panels.	Increases in the system's electrical efficiency by 7.17 %	NPCM are useful temperature controllers for PV solar cells.
[92]	2022	paraffin PCM with expanded graphite and nano-copper	Experimental and numerical	The melting temperature of the paraffine PCM was 62–64 °C. The expanded graphite possessed a 80 mesh size, a 250 mL/g expansion rate, and an identified surface area of around 40 m2/g. Cu nanoparticles measured 100 nm in size.	Cu considerably facilitated the phase-change operation of the composite PCMs, increasing the heat-response rates of storage and release by 152.17 % and 100 %, accordingly.	The efficiency of the composite PCMs increased greatly when compared with pure paraffin PCM, demonstrating that such composite PCMs are suited for solar passive heating.
[93]	2022	nano-graphite/paraffin composite PCM	Experimental	A finned tube-heat exchanger is put into the PCM to delay its melting. In the first phase, a PV panel is cooled using pure paraffin and a finned tube-heat exchanger, and then graphite nanoparticles (= 0.002–0.012 (w/v)) are coupled with PCM.	By employing nano-graphite PCM with = 0.01 (w/v) and the greatest amount of water flow rate (Q = 100 mL s1), the mean temperature at the surface of the PV module is decreased from 336.15 to 310.25 K. The largest increase in power production of the PV/nano-PCM combination achieved by adopting efficient parameters is 21.2 %.	The suggested technology decreases the photovoltaic panel's temperature while maintaining it consistent for each cell.
[94]	2021	PT-58 (PureTemp) + multiwall carbon nanotubes	Experimental	PCM integration on the back side of the panel,	electrical power increased by 32.29 %	Cooling PV panels using different hybrid nanoparticles PCMs can improve their efficiency.
[94]	2021	PT-58 (PureTemp) + Graphene nanoplatelets	Experimental	PCM integration on the back side of the panel,	electrical power increased by 33.07 %	In comparison to previous systems, the current study enhanced the efficiency of PV panels. The use of nano-PCMs was successful in cooling PV panels.
[94]	2021	PT-58 (PureTemp) + magnesium oxide (MgO)	Experimental	PCM integration on the back side of the panel,	electrical power increased by 30.45 %	The current system boosted electrical power as compared to standard panels, and electrical energy was also increased when PV panels combined with micro PCMs were used.
[95]	2021	Paraffin with 0.5 % volume of TiO2	Experimental	In this work, two solar PV panels with comparable configurations were tested, one without any nano PCM and the other with nano PCM incorporated behind the panel.	The average daily surface temperature of the panel was reduced by 13 °C, and the panel's daily efficiency was increased by 2.1 %.	The findings demonstrated that the NPCM was successfully beneficial

Table 7
Recent studies ranging between 2020 and 2023 on cooling solar systems using finned phase change materials, with the method of study, system configuration and setup, results obtained and a final conclusion.

Refs.	Year	PCM Used	Methods	Configuration	Results	Conclusion and notes
[97]	2023	RT 35-HC (Latent heat 240,000 J/kg)	Software simulation (Ansys fluent 15.0 CFD software)	PCM (RT35HC) is installed with a series of fins made of various materials (graphite, copper, steel, and titanium). The simulation is performed at various inclination degrees (0, 30, 45, 75, and 90°).	When graphite and copper were used instead of steel and titanium, the average temperature of the solar panel was lowered by 3 °C. The total temperature was regulated at roughly 40 °C.	Improvement in system performance, verified through previous experimental studies
[98]	2023	Paraffin (RT42) with aluminum fins	Software simulation (Ansys fluent and C++)	The model is made up of a PCM layers with an aluminum rectangular fin installed on the rear side of the panel.	The temperature is reduced by 32 K. Furthermore, it have shown that raising PCM thickness reduces the PV temperature even more.	Lower thicknesses dissolve fast in the daytime sunshine, but larger thicknesses do not melt completely.
[90]	2023	Low-cost aluminum reflectors, sinks, and Paraffin was PCM combined with oxide of zinc (ZnO) nanoparticles	Experimental	Adding a low-cost aluminum reflector and incorporating PCM (Paraffin wax)/zinc oxide (ZnO) nanoparticles. Outdoor climatic circumstances at Latitude: 56.84°N, Longitude: 60.64°E, Delta-SM-12 M photovoltaic panel with capacity of 50 W was employed for the investigation.	Reduction of 28.3 %. in temperature and increase in electrical power output of 18.16 %	Fins with PCM and nanoparticles results in higher panels efficiencies due to reduced temperatures
[99]	2023	Inorganic PCM made of salt and water ($CaCl_2{\cdot}6H_2O$) and copper fins	Numerical	The solar collector has a 30° inclination angle and is made of a plexiglass shelter, the panel, copper sheet, and plexiglass containing an insulating layer. (11 Fins used)	Average temperature 48.58 °C instead of 64.69 °C, as well as higher efficiency by 11.19 %.	The varied size and nonuniform configuration of fins may improve PV cell performance even further.
[100]	2023	Expanded graphite form stable PCM of polyethylene glycol	Experimental	During the melting and cooling of the panel, this system maintained the form and maintained adequate thermal interface with the panel and PCM	Enhanced PV panel efficiency of 3.667 %, which is greater than the typical cooling technology (heat sink), which is 1.072 %.	The system can cool solar PV panels in a temperature range from 35 to 40 °C, increasing its power production, efficiency, and life duration.
[85]	2021	Paraffin with fins	Experimental	melting temperatures ranging 56–58 °C. PCM is colorless, non-toxic, odorless, and non-acidic. placed on the backside of 60 W PV panel	Temperature drops by 47 °C and efficiency increase by 5.94 %	The use of PCM and the placement of fins on the backside of the panels improves the overall efficiency of the PV modules.
[85]	2021	Polyethylene glycol 600 (PEG-600) with fins	Experimental	PEG-600 melting temperatures ranging from 23 to 26 °C, it is colorless, non-toxic, odorless, and non-acidic. And placed on the backside of 60 W PV panel.	Temperature drops by 34 °C and efficiency increase by 4.65 %	The use of PCM and the placement of fins on the backside of the panels improves the overall efficiency of the PV modules.
[101]	2021	Paraffin wax and aluminum fins	Experimental	The experiment was carried out in July during the summer season, with PCM attached to the back side of the panel.	Temperature was reduced by 12.13 °C, produced power was increased by 11.33 %, and electrical efficiency was increased by 5.15 %.	The suggested cooling system may be used in hot weather circumstances to improve the efficiency of solar photovoltaic cells at affordable costs.
[102]	2020	Palm wax with finned box	Experimental	A 20 W polycrystalline PV module was employed, with the PCM box connected to the back surface of the PV, at 15° inclination towards south.	a 6.1 °C temperature decrease and a 5.3 % increase in module efficiency were recorded	Palm wax was revealed to have promised as a PCM able to cool the PV module.

Table 8
Recent studies ranging between 2020 and 2023 on cooling solar systems using hybrid systems with phase change materials, including the method of study, system configuration and setup, results obtained and a final conclusion.

Refs.	Year	PCM Used	Methods	Configuration	Results	Conclusion and notes
[105]	2023	Encapsulated PCM balls	Experimental	Employing active and passive cooling methods to compare between them at 3 l/min cooling water flow rate	The PVT-PCM panel cooled by the active method produced the greatest electrical and thermal energy gain, yielding an overall efficiency of 74.1 % against 34.6 % for passive cooling method	To satisfy the demands needed for practical applications, the methods used systems accomplish a larger electricity production and energy storage capacity with a reduced frictional power.
[106]	2023	Nano PCM (SiC 1 % Vol) and Nanofluid (SiC 0.6% Vol)	Experimental	A controlled environment employing an indoor solar simulator, placing the cooling nanofluid flow system with a micro fin pipe (inner grooved) encased with nano PCM	The temperature dropped by almost 45 °C. The suggested system achieved 77.5 % greater thermal efficiency than a standard circular tube without micro-fins. Furthermore, the system generated power is 14.5 W, instead of 10.49 W for the PV panel.	Because to the use of nanoparticles, nano PCM and nanofluid have greater thermal conductivities, enhancing their efficiency.
[107]	2022	SP31 with heat pipes	Numerical (MATLAB)	3 cm thickness tested in summer	temperature reduction of 20.6 °C and electrical efficiency improvement of 11.5 %	PCM should be chosen based on weather conditions
[107]	2022	SP15-gel With heat pipes	Numerical (MATLAB)	2.5 cm thickness tested in winter conditions	attaining 14.1 °C temperature decrease with 7.2 % increase in PV electrical efficiency.	PCM should be chosen based on weather conditions
[108]	2022	Phase change materials integrated with porous medium made of aluminum	Experimental	2 cm thick porous medium	temperature drop of 21 °C	The use of two passive cooling systems is combined and resulted in better results
[109]	2022	Vaseline PCMs, Aluminum Heat Sinks , thermoelectric, silver back casing	Software Simulation (COMSOL Multiphysics software)	The research is related to a 10.5 W polycrystalline Photovoltaic Panel.	The largest increases in electrical performance and panel efficiency are 19.4 % and 19.32 %, respectively.	The combination of the Peltier effect with latent energy storage enables increased cooling to improve the performance of solar modules.
[110]	2022	paraffin wax (PCM) -filled aluminum cans	Experimental	Improved and unimproved PV panels were tested consecutively in order to examine and compare their performance.	The electricity efficiency has raised from 10.69 % till 12.60 %. Furthermore, the average electric power efficiency was determined to be 9.84 % higher.	The use of paraffin wax-filled cans of aluminum as a heat management and efficiency enhancement technology in solar systems was effective.
[111]	2022	Aluminum metal foam embedded with RT-44 (RUBITHERM) PCM	Experimental	SM-20P and 20 W PV Panel was designed and constructed at the rooftop of a building at north latitude 30.466 North and longitude 31.185	Higher electricity generation than typical by 1.85 %, 3.38 %, and 4.14 % in December, January, and February, respectively, with temperatures 4 %, 7.4 %, and 13.2 % lower.	The use of the PCM types and their effects highly depends on the time of the year
[112]	2022	Two types of different properties enhanced PCM's with thermoelectric generators	Experimental	A photovoltaic (PV) panel, two PCM materials with distinct phase transition temperatures, and an additional layer of TEG modules in between the PCMs were used in the design.	The mean solar cell temperature was reduced by 89.4 %, while the thermoelectric generator gradient of temperature was increased by 1.2 %. The system's mean electrical energy and total exergy efficiencies were 8.1 % and 9.9 % higher, respectively, than without PCM	Metallic thermal transfer enhancers boost the heat transfer rate of PCMs. Two PCMs maintain the temperature differential for the TEGs, while the top PCM feeds low-temperature heat energy through the water-carrying tubes and decreases the PV surface temperature.
[113]	2021	RT-25 with thermos electric generators	Numerical	A thermoelectric generator and phase change material are used in a hybrid cooling system for solar panel heat dissipation.	Panel efficiency improves by 2.5 % in favorable circumstances and 3.5 % in sunny conditions. In addition, in favorable conditions, electrical power generation increases by 20 %, and in bright weather, it increases by 30 %.	Even under transitory irradiance flux, the solar panel's efficiency improvement is consistent over 10 h.

(*continued on next page*)

Table 8 (*continued*)

Refs.	Year	PCM Used	Methods	Configuration	Results	Conclusion and notes
[114]	2021	carbon nanotube additives (concentration ratio of 0.2 %) and completely refined paraffin wax PCM and a copper foam matrix	Experimental	PV panel performance was investigated experimentally in a case study of a hot climate zone.	increase in average PV panel electrical efficiency from 4 % to 21 %	Through heat absorption and rejection, porous metals significantly increased the material's effective thermal characteristics.
[115]	2021	PVT with RUBITHERM, RT-35 PCM	Experimental	The functioning of a hybrid PVT-PCM panel and a standard photovoltaic (PV) module at the same time. Both photovoltaic modules utilized are the same brand, model, and technical specs.	The hybrid PVT-PCM's daily electricity efficiency improved by 7.43 % when compared to the standard PV module. Furthermore, the PVT-PCM module's daily hybrid efficiency is 31.35 %, while the PV module's daily electrical efficiency is 13.12 %.	A lower and more constant operating temperature is obtained by using a phase change material within a combination PVT-PCM.
[116]	2020	Calcium chloride hexahydrate ($CaCl_2{\cdot}6H_2O$) with aluminum containers	Experimental	The panel was exposed to the sun outside. The experiment was carried out on clear weather days on the 26th of January 2019 and the 15th of June 2019 from 7:00 a.m. to 6:00 p.m. Panel was sloped at a 13° angle facing south.	The maximum PV temperature fell from 64.4 °C to 46.4 °C in January and from 77.1 °C to 53.8 °C in June. The electricity efficiency improved from 9.5 % to 10.5 % throughout the middle of the day. Daily power generation rose up by 6.2 % in January and 8.3 % in June.	The heat may be employed for water heating by including a heat exchanger within the PCM containers, as well as for space heating in cold areas. It will improve the whole performance of the system by increasing useable heat output as well as daily power generation.
[86]	2020	PVT- PCM employing RT-35HC (RUBITHERM) - Organic Paraffin PCM and Graphene nanoparticles of nanofluid	Experimental	Experiments were carried out from 9 a.m. to 4 p.m. on days with almost identical ambient conditions in July. 30 W monocrystalline PV panels were utilized. To gather the most sunrays, the tilt angle of the solar panel was set at around 34° (latitude angle: 33.7).	Temperature dropped by 23.9 °C and electrical efficiency increased by 23.9 %	The usage of nanofluid in conjunction with PCM improves system performance over PCM alone. Furthermore, a combination system of PCM and nanofluid is more effective than the integrated PCM and water system.
[86]	2020	PVT- PCM employing RT-35HC (RUBITHERM) - Organic Paraffin PCM with water	Experimental	Experiments were carried out from 9 a.m. to 4 p.m. on days with almost identical ambient conditions in July. 30 W monocrystalline PV panels were utilized. To gather the most sunrays, the tilt angle of the solar panel was set at around 34° (latitude angle: 33.7).	Temperature dropped by 16.1 °C and electrical efficiency increased by 22.7 %	The usage of nanofluid in conjunction with PCM improves system performance over PCM alone. Furthermore, a combination system of PCM and nanofluid is more effective than the integrated PCM and water system.
[117]	2020	organic fully refined paraffin wax (C27H56)/copper foam matrix and 0.20 % multi-walled carbon nanotubes	Experimental	A single square PV cell with an area of 182.25 cm2 was utilized in the experiment, and the cell was combined with its PCM. The PV cell is mono-crystalline, and its maximum power output under normal circumstances (1000 W/m2 at 25 C) is 1.4 W.	The temperature of the cell is lowered by 13.29 % while its efficiency of producing electricity increased by 5.68 %.	A copper foam matrix containing carbon nanotubes with multiple walls has the potential to be employed as a natural cooling solution to efficiently manage the temperature of a solar cell.

ficiency. It created a computational framework for thermal evaluations of the integrated system, which was validated using experimental data. A 50 mm thick PCM layer with a thermal conductivity of 5 W/m K and a phase transition temperature of 40–45 °C provided the optimum thermal efficiency for the PV/TEG/PCM system. However, more optimization and experimental assessment are advised in order to identify the complete technical and scientific bounds [104].

4. Discussion and analysis

4.1. Results analysis and discussion

After gathering and analyzing all the needed articles and previous studies about cooling solar photovoltaic systems using phase change materials, each type of cooling method, which are:

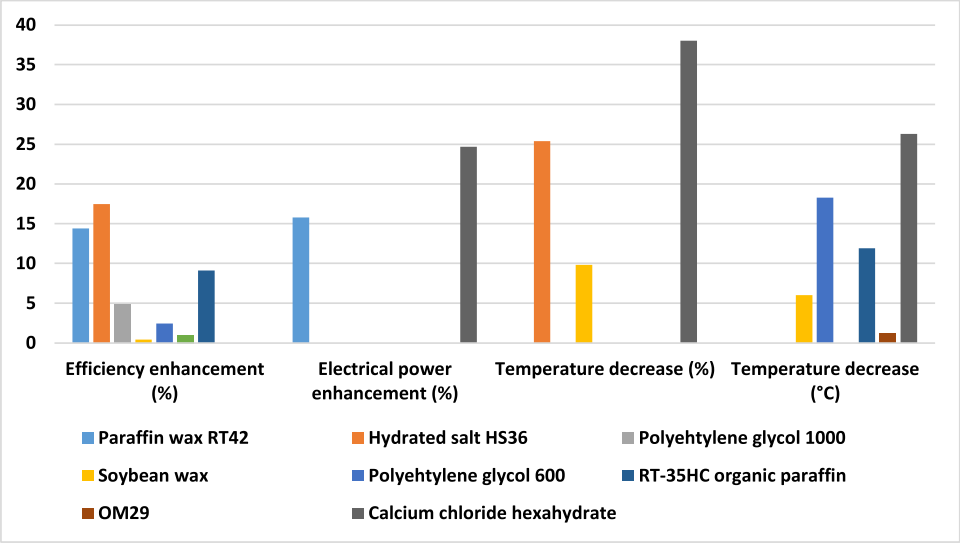


Fig. 12. The different types of pure PCM cooling systems and their effects on the solar system.

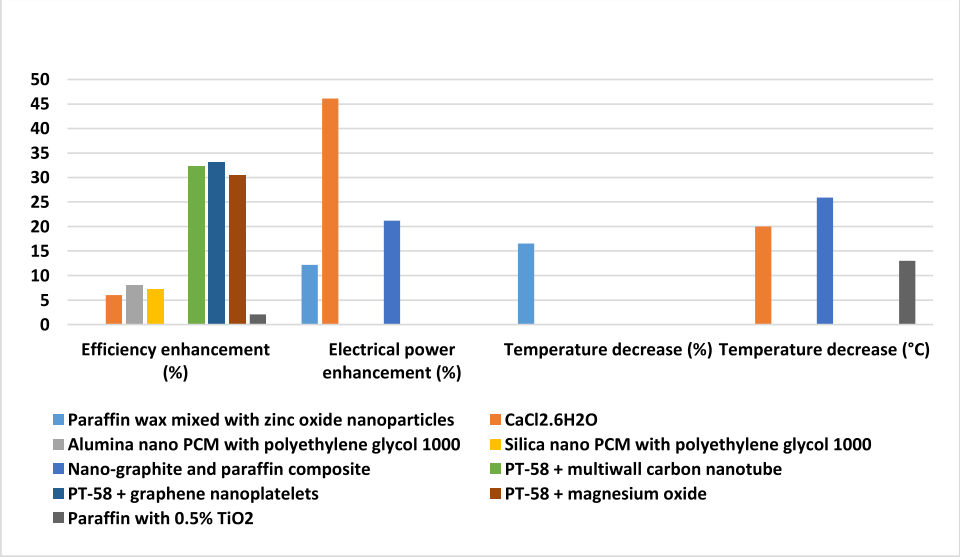


Fig. 13. The different types of Composite PCM cooling systems and their effects on the solar system.

- Pure/Simple PCM
- Composite PCM
- Finned PCM
- Hybrid cooling systems, including PCM

was filtered alone with its own research papers and their corresponding improvements to the system in order to have the proper discussion and analysis process and get beneficial results throughout this research. Where the papers are classified based on the PCMs used in each method and also based on their provided data, the outcomes of the research on the enhancement and the effect of cooling in the system could be classified into four types of outcomes, which are:

- Efficiency enhancement (%)
- Electrical power enhancement (%)
- Temperature decrease (%)
- Temperature decrease (˚C)

Based on these outcomes, systems are to be compared, analyzed, and discussed.

Concerning the pure PCM, as shown in the barograph in Fig. 12, when talking about efficiency enhancement, the hydrated salt HS36 had the highest effect on increasing the efficiency of the system by 17.5 %, and the paraffin wax RT42 PCM had a remarkable efficiency enhancement of 14.4 %. However, when moving to the electrical power enhancement and the percentage of decrease in the temperature, calcium chloride hexahydrate had the highest values of 24.68 % and 38 %, respectively. The lowest efficiency enhancement was obtained by the soybean wax PCM of 0.42 %. As a conclusion, paraffin wax RT42 and hydrated salt HS36 can be employed as pure PCM to enhance solar system efficiency. Moreover, more studies might be conducted on the calcium chloride hexahydrate to determine its effect on the system efficiency and use it as a pure cooling PCM for solar cells.

Moving to the composite PCM use in cooling systems, as shown in Fig. 13, the highest efficiency enhancements were too close for the three systems including PT-58 PCM however each time it was integrated with a different type of composite. Where the first time with multiwall carbon, the second time with graphene nanoplatelets and the third time with magnesium oxide, and they recorded efficiency enhancements by 32.29 %, 33.07 % and 30.45 % respectively. Where the highest electrical power improvement was by employing $CaCl_2{\cdot}6H_2O$ by 46.11 %. The use of Paraffin with 0.5 % TiO2 wasn't beneficial, since it only resulted in 2.1 % efficiency enhancement.

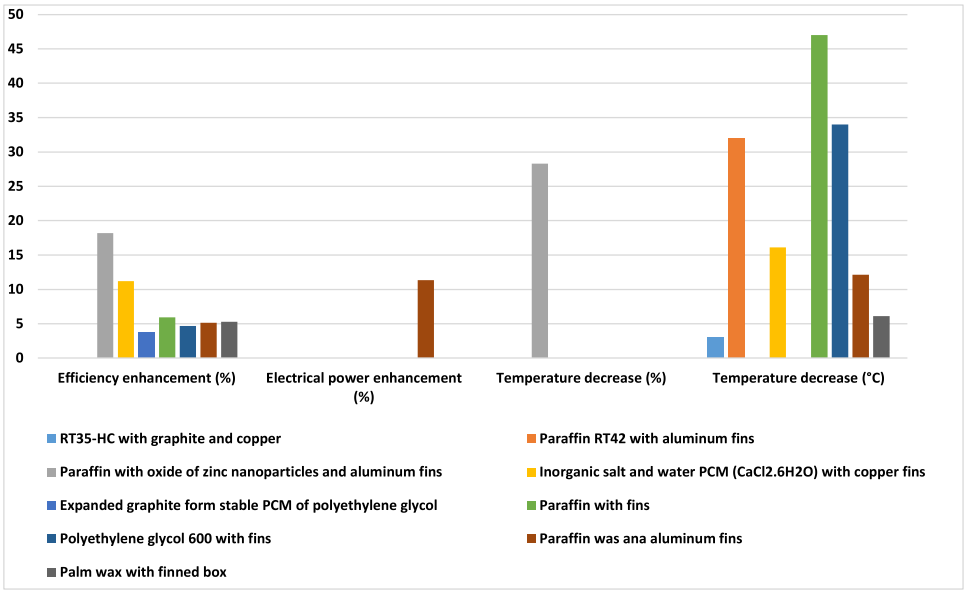


Fig. 14. The different types of composite PCM cooling systems and their effects on the solar system.

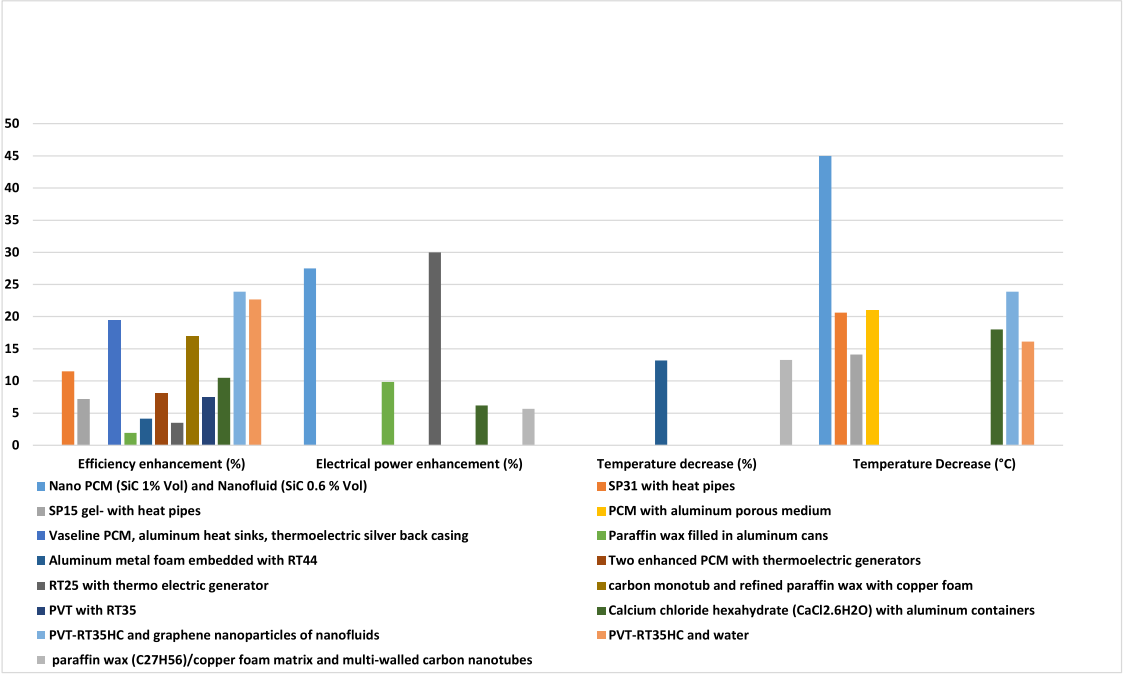


Fig. 15. The different types of Hybrids PCM cooling systems and their effects on the solar system.

After investigating pure and composite PCM, the effect of finned PCM was also investigated.

The highest efficiency enhancements were achieved by paraffin with oxides of zinc nanoparticles and aluminum fins, and by inorganic salt and water PCM ($CaCl_2{\cdot}6H_2O$) with respectively values of 18.16 % and 11.19 %. The rest of the cooling systems using finned PCM had almost-close efficiency improvements whose values ranged between 3 and 6% as shown in Fig. 14.

Finally, moving to the Hybrid cooling systems integrated with PCM shown in Fig. 15, the highest efficiency improvements values were obtained using integrated PVT-RT35HC with graphene nanoparticles of nanofluids, and with water, where the efficiency enhancement was recorded to be 23.9 % and 22.7 % . Close to these values is the enhancement of 19.4 % obtained by Vaseline PCM integrated with aluminum heat sinks and thermoelectric silver back casing. After it also high efficiency increasing is the carbon monotube and refined paraffin wax with copper foam with 17 % enhancement. However, the rest of the values ranges between 3.5 % and 11.5% improvements. Moreover, it is worth to mention that the use of Nano PCM (SiC 1 % Vol) and Nanofluid (SiC 0.6 % Vol) resulted in 27.5 % electrical power enhancement and 45 °C temperature reduction.

4.2. *Different systems and publications dates discussion analysis*

As shows Fig. 16, the research on finned PCM and hybrid PCM were increasing gradually over the past years specifically the last two years during 2023 and 2022. However, pure PCM research studies were constant and slightly decreasing with time and years, as same for the composite PCM researches, where they started during the last year to decrease slightly. It is recommended based on the literature review men-

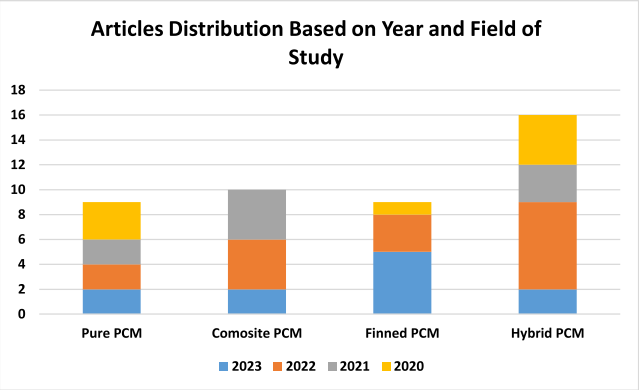


Fig. 16. Articles distribution based on the year and the field of study.

tioned in this paper and based on Fig. 16, to investigate more about PCM applications, particularly hybrid and finned cooling systems, since they also had high values in improving the cell's efficiency and electrical power output.

5. Main challenges for using PCM as PV cooling solution

The installation and maintenance techniques employed in the recently mentioned studies that employ Phase Change Materials to enhance the thermal management of photovoltaic systems are the main subject of Tables 5–8. The methods employed to accomplish different PCM applications are examined; these methods include PCM based on nanoparticles [121], hybrid PVT/PCM systems [120], enclosed PCM balls [118], complex combinations with thermoelectric generators [119], etc. Additionally, it draws attention to suggested maintenance concerns, highlighting the necessity of ensuring the long-term stability and ongoing effectiveness of advanced solar energy technologies [122].

Attaching Phase Change Materials (PCMs) to the back surface of a photovoltaic (PV) panel is one recommended installation method [123], which takes into account the PCM thickness and tilt angle [124]. On the other hand, regular inspections are necessary to guarantee peak performance. It is essential to routinely check the tilt angle in order to preserve efficiency [125]. In addition, if a PCM container is used on the PV panel's back plate, it is necessary to regularly check the PCM containment system for possible leaks or deterioration. To maintain the overall functioning of the system, structural integrity examinations of the PCM container should also be carried out on a regular basis.

Periodic examinations are crucial for composite phase change material systems [126] in order to identify any indications of degradation or separation within the composite material. This guarantees that the composite PCM will continue to improve PV panel thermal management.

A regular inspection schedule is required when PCM systems are integrated with fins made of different materials [127] in order to spot any corrosion or damage in the fins. Frequent inspections help the fins remain long and continue to function, which is essential for effective heat dissipation.

6. Conclusions and recommendations

6.1. Conclusions

- The global demand for renewable energy sources, specifically solar systems, is increasing with the years due to climate change and energy sustainability concerns.
- Solar systems are subjected to overheating that can decrease their efficiency and durability, which requires effective cooling solutions.
- Recent research focused on implementing phase change materials (PCMs) to solve the overheating issue in solar system, resulting in four distinct cooling strategies: pure PCM, composite PCM, finned PCM, and hybrid PCM systems.
- The choice of PCM has an important role in determining the effectiveness of cooling systems, which was shown when using calcium chloride hexahydrate and Paraffin Wax RT42 as pure PCM; notable efficiency enhancements of up to 17.5 % were remarked.
- Composite PCMs, specifically when integrated with multiwall carbon, graphene nanoplatelets, and magnesium oxide, have enhanced cooling systems efficiency by exceeding 30 %.
- Finned PCM systems using zinc nanoparticles and aluminum fins have cost-effectiveness and high performance when used for PV cooling, with gains up to 18.16 %.
- The integration of PCMs with various technologies resulting in hybrid cooling systems, such as thermoelectric generators and nanofluids, holds significant efficiency improvements of up to 27.5 %.
- Research trends indicate a growing interest in finned and hybrid PCM systems, with studies peaking in 2022 and 2023.

6.2. Recommendations and future work

- Conduct comprehensive research to identify the PCM types that offer the highest thermal performance (broader phase-change temperature range, increased thermal conductivity, and improved thermal stability), durability, and cost-effectiveness for cooling PV panels.
- Integrate PCM-based cooling systems with energy storage technologies (batteries or super capacitors), creating energy management solutions that enhance grid stability and reliability.
- Perform comprehensive economic analyses (life cycle cost assessments) to determine the economic feasibility and return on investment for PCM-based cooling systems.
- Conduct lifecycle assessments (LCA) to evaluate environmental impact of PCM materials (production, use and disposal) to ensure that PCM-based cooling solutions align with sustainability goals.
- Apply machine learning and artificial intelligence techniques to optimize PCM cooling operation leveraging historical performance data and weather forecasts for predictive control strategies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Razan El Kassar: Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Ahmad Al Takash:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Jalal Faraj:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Writing – review & editing. **Mahmoud Khaled:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Haitham S. Ramadan:** Conceptualization, Formal analysis, Investigation, Writing – review & editing.

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